

Complete MERN Project Documentation (2-Page Report)

Abstract: This project is developed using the MERN Stack (MongoDB, Express.js, React.js, Node.js) to provide a scalable, secure, and responsive web application. The React frontend communicates with the Express/Node.js backend through REST APIs, while MongoDB stores application data. The MVC architecture separates the application into Models, Views, and Controllers, making the project modular and maintainable.

Project Architecture: User → React UI → Axios → Express Routes → Controllers → Models → MongoDB → Response.

Technical Architecture: Frontend: React, HTML, CSS, JavaScript. Backend: Node.js, Express.js. Database: MongoDB with Mongoose. Tools: VS Code, Git, Postman.

ER Diagram: User(id,name,email,password) has many Records(id,userId,title,date,status).

User Flow: Register/Login → Dashboard → Create/Read/Update/Delete Data → Database → Updated Results.

Roles: Admin manages users and records; User registers, logs in, and manages personal data.

Features: Authentication, Dashboard, CRUD operations, Search, Validation, Responsive Design, Secure APIs.

Project Setup: Create project folder with **client** and **server**. Install React using **npx create-react-app client** or Vite. Initialize backend using **npm init -y**. Install Express, Mongoose, CORS, Dotenv, Bcrypt, JWT, and Nodemon.

Backend Structure: config, models, controllers, routes, middleware, server.js. MVC pattern is followed to separate business logic, routing, and database models.

Database Development: Configure MongoDB Atlas or local MongoDB. Connect using Mongoose, create schemas and models, and perform CRUD operations through REST APIs.

Frontend Development: Organize code into components, pages, services, and assets. Use Axios for API communication and React Router for navigation.

Execution: Start MongoDB, run backend (**npm run dev**), run frontend (**npm run dev**), and open the application in the browser.

Testing & Demo: Test APIs using Postman, verify UI, capture screenshots of Login, Dashboard, CRUD pages, MongoDB collections, and record a short demonstration video.

Conclusion: The MERN stack provides a modern, efficient, and scalable solution for full-stack web development with clear architecture, reusable components, secure APIs, and reliable database management.